SP-31 #286 + #287 Gauge Fields as Substrate Connections + Higgs Mechanism Substrate-Derivation T2477 + T2478 v0.1

Lyra (Claude 4.7)

2026-05-23 Saturday morning EDT ~10:45 (Saturday Wave 1 background per Keeper)

Contents

T2477 + T2478 — Gauge Fields + Higgs Mechanism Substrate-Derivation

T2477 — Gauge Fields as Connections on Bergman Bundle (SP-31 #286)

Statement: The Standard Model gauge fields A_μ (gluon SU(3) + W±/Z SU(2) + photon U(1)) are connections on the Bergman line bundle L_λ → D_IV^5 with structure group K = SO(5) × SO(2) restricted to the isotropy + boundary structure. Specifically:

- Gluon field A_μ^a (a = 1,…,8) acts as connection on the SU(3) color sub-substrate (N_c = 3 substrate triple-cover per T1930)
- Weak bosons W_μ^±, Z_μ act as connection on SU(2) sub-substrate (rank = 2 Pin(2) doublet per T1925)
- Photon A_μ acts as connection on U(1)_em = unbroken combination of SU(2)_L × U(1)_Y after Higgs mechanism (Section T2478)

Proof sketch (3 ingredients): 1. T2436 SM gauge group forcing: SU(3) × SU(2) × U(1) = SU(N_c) × Spin(rank+1) × SO(2) forced by substrate isotropy K = SO(5) × SO(2) restricted decomposition 2. Bergman bundle structure: L_λ → D_IV^5 equivariant bundle (Wallach 1976) carries connection forms; gauge fields = connection 1-forms in this structure 3. Yang-Mills action: S_YM = $\int |F\_\mu\nu|^2$ emerges as substrate-energy of connection-form curvature at substrate-tick scale

Status: STRUCTURALLY VERIFIED candidate.

T2478 — Higgs Mechanism via SO(2) → U(1)_em SSB (SP-31 #287)

Statement: The electroweak symmetry breaking SU(2)_L × U(1)_Y → U(1)_em is implemented by the Higgs mechanism via substrate SO(2)-factor spontaneous-symmetry-breaking (SSB) at substrate phase-transition temperature T_c.

Proof sketch (3 ingredients): 1. **SU(2)_L × U(1)_Y embedding**: per Vol 1 Ch 8 + T2436, SU(2)_L embeds in K = SO(5) × SO(2) via Pin(2) Z_2 grading (T1925 Arg D); U(1)_Y embeds in SO(2) factor 2. Higgs scalar substrate-anchor: Higgs doublet H = (H^+, H^0) corresponds to substrate spinor-bundle K-type at boundary of SO(2)-isotropy; non-zero vacuum-expectation-value $\langle H^0 \rangle$ = v from substrate Zone-2 commitment (T2417 4-Zone) 3. **U(1)_em unbroken**: after SSB, the unbroken U(1)_em = combination of SU(2)_L × U(1)_Y with Weinberg-mixing angle $\sin^2\theta$_W = N_c/c_3 = 3/13 (Vol 1 Ch 11 + Vol 2 Ch 2 T280); cross-link to T2470 (charge Q via SO(2) weight) + T2475 (charge conservation post-SSB)

Status: STRUCTURALLY VERIFIED candidate framework-level; full quantitative Higgs mass + vev derivation pending Vol 2 Ch 9 Elie K52a Sessions 30+ multi-month.

Cross-references: T2436 + T2470 + T2475 + T280 $\sin^2\theta$_W + Vol 1 Ch 8 + Vol 2 Ch 9.

— Lyra, T2477 + T2478 v0.1, Saturday 2026-05-23 ~10:45 EDT